When researching such issues, ecologists have tended to use both traditional observational fieldwork and also sophisticated mathematical models. More recently, research has begun to incorporate the manipulation of ecosystems in a more controlled way, on plots of land, for example, or within closed systems housed in giant greenhouselike facilities called phytotrons. The experiments help to establish what factors—such as numbers of species, or species type and dominance—affect ecosystems in the long term. Their findings show that the effects of biodiversity on ecosystem functions are complex. While the most diverse ecosystems tend to be the most productive, their success also depends on climate and soil fertility.

There is more to be learned about how plant diversity affects soil processes, the role of microbe biodiversity in the soil, and the effects of mutualistic species, such as flowering plants and pollinating insects. Much has been achieved, but questions remain, and the unifying theory that Loreau is seeking has still to be devised.